

AI text detector · full pilot comparison

AI recall = generated texts detected · Human FPR = human texts falsely flagged · Dashed marker = 2% FPR target

MODEL / TRAIN TIER	AI RECALL · 0-100%		HUMAN FALSE POSITIVES	
MIXED TEST				
631 human + 631 AI · 35% papers in each class				
Qwen3 LoRA · ours		100.0% (631/631)		3.0% (19/631)
Character TF-IDF · full		97.0% (612/631)		2.4% (15/631)
Character TF-IDF · medium		95.2% (601/631)		2.1% (13/631)
Character TF-IDF · small		89.7% (566/631)		2.7% (17/631)
Character TF-IDF · tiny		67.2% (424/631)		1.3% (8/631)
Word TF-IDF · medium		90.8% (573/631)		1.9% (12/631)
MiniLM embedding · medium		30.3% (191/631)		1.4% (9/631)
EditLens RoBERTa		97.8% (617/631)		1.3% (8/631)
EditLens Llama		99.7% (629/631)		2.5% (16/631)
PAPER ABSTRACT TEST				
221 human + 221 AI · held-out venues				
Qwen3 LoRA · mixed-trained		100.0% (221/221)		4.5% (10/221)
Character TF-IDF · full		99.1% (219/221)		3.2% (7/221)
Character TF-IDF · medium		99.1% (219/221)		2.7% (6/221)
Character TF-IDF · small		98.6% (218/221)		2.7% (6/221)
Character TF-IDF · tiny		95.9% (212/221)		2.7% (6/221)
Word TF-IDF · full		99.1% (219/221)		2.3% (5/221)
MiniLM embedding · full		58.4% (129/221)		2.7% (6/221)
EditLens RoBERTa		91.9% (203/221)		0.9% (2/221)
EditLens Llama		98.6% (218/221)		1.4% (3/221)
GENERAL EDITLENS TEST				
2,000 human + 2,000 AI; Llama scored 500 + 500				
Character TF-IDF · medium		98.9% (1978/2000)		1.9% (39/2000)
Word TF-IDF · medium		96.4% (1928/2000)		2.0% (40/2000)
MiniLM embedding · medium		47.1% (942/2000)		1.6% (31/2000)
EditLens RoBERTa		100.0% (1999/2000)		3.1% (62/2000)
EditLens Llama · small test		100.0% (500/500)		3.6% (18/500)
ENRON EMAIL TEST				
1,800 human + 1,800 AI; Llama scored 500 + 500				
Character TF-IDF · general medium		75.9% (1366/1800)		0.4% (7/1800)
Word TF-IDF · general medium		71.0% (1278/1800)		2.1% (38/1800)
MiniLM embedding · general medium		31.7% (571/1800)		0.9% (17/1800)
EditLens RoBERTa		98.1% (1766/1800)		2.7% (49/1800)
EditLens Llama · small test		100.0% (500/500)		1.0% (5/500)
PMC ABSTRACT TEST				
87 human + 87 AI · held-out journals				
Character TF-IDF · full		96.6% (84/87)		0.0% (0/87)
Word TF-IDF · full		97.7% (85/87)		2.3% (2/87)
MiniLM embedding · full		20.7% (18/87)		1.1% (1/87)
EditLens RoBERTa		92.0% (80/87)		1.1% (1/87)
EditLens Llama		97.7% (85/87)		1.1% (1/87)
SWAPPED-GENERATOR PAPER TEST				
182 human + 182 AI · same held-out human works, alternate generator				
Qwen3 LoRA · ours		100.0% (182/182)		3.3% (6/182)
Character TF-IDF · paper full		97.3% (177/182)		2.7% (5/182)
Word TF-IDF · paper full		100.0% (182/182)		2.7% (5/182)
MiniLM embedding · paper full		61.0% (111/182)		2.2% (4/182)
EditLens RoBERTa		90.1% (164/182)		1.1% (2/182)
EditLens Llama		100.0% (182/182)		1.1% (2/182)
HUMAN ACL ABSTRACT AUDIT				
human only · paper/mixed audits exclude sampled paper works; general audits have a larger pool				
Character TF-IDF · mixed medium		human only		1.8% (317/17560)
Word TF-IDF · mixed medium		human only		0.8% (139/17560)
Character TF-IDF · paper full		human only		1.0% (177/17560)
Character TF-IDF · general only		human only		64.8% (11854/18287)
Word TF-IDF · general only		human only		49.1% (8983/18287)
MiniLM embedding · general only		human only		1.4% (259/18287)
HUMAN PMC BODY AUDIT				
261 human-only text chunks from held-out full papers · false positives only				
Qwen3 LoRA · ours		human only		5.7% (15/261)
Character TF-IDF · PMC full		human only		2.7% (7/261)
EditLens RoBERTa · PMC threshold		human only		0.8% (2/261)
Character TF-IDF · general only		human only		34.9% (91/261)

FPR bars use a 0–10% scale; ACL and PMC body audits use 0–70% and 0–40%. Percentages are observed rates, not confidence bounds. Thresholds were selected on each model's validation set. The Qwen3 paper subset uses its mixed-set threshold. Reference checkpoints had larger external training data. Paper AI is title-conditioned text from two small local generators; the swapped test reuses the same held-out human works. EditLens assets are noncommercial.